

BSs Computer Science
Probability and statistics – 2nd semester - Mid term
April 29, 2026
Duration: 2 hours • Total: 22 points

Use of phones, documents, or any form of cheating is strictly prohibited.
Any student who caught cheating will receive a grade of 0.

Exercise 1 – Multiple Choice Question (2.5 pts)

Each question has exactly one correct answer. No penalty for wrong answers.

Question 1 (0.5 pt)

Which of the following is NOT an assumption of one-way ANOVA?

- A) The observations within each group are independent.
- B) The variances of the groups are approximately equal.
- C) The dependent variable follows a normal distribution within each group.
- D) The sample sizes of all groups must be exactly equal.

Question 2 (0.5 pt)

A 95% confidence interval for a population mean is computed as [12.4;18.6]. Which of the following statements is correct?

- A) There is a 95% probability that the true population mean lies between 12.4 and 18.6.
- B) If we repeat the sampling procedure many times, approximately 95% of the constructed intervals will contain the true mean.
- C) 95% of the observed data values fall between 12.4 and 18.6.
- D) The sample mean is equal to 15.5 with probability 0.95.

Question 3 (0.5 pt)

The coefficient of determination R^2 equals 0.82 for a simple linear regression. Which interpretation is correct?

- A) The slope of the regression line is 0.82.
- B) 82% of the variability in Y is explained by the linear relationship with X.
- C) The correlation coefficient r is equal to 0.82.
- D) The model makes prediction errors of at most 18%.

Question 4 (0.5 pt)

In a multiple linear regression model with two predictors x_1 and x_2 , the coefficient β_1 represents:

- A) The expected change in Y when x_1 increases by one unit, ignoring x_2 .
- B) The expected change in Y when x_1 increases by one unit, holding x_2 constant.
- C) The correlation between x_1 and Y.
- D) The proportion of variance in Y explained by x_1 alone.

Question 5 (0.5 pt)

A researcher wants to include a categorical variable 'Season' (Spring, Summer, Autumn, Winter) in a regression model using dummy variables. How many dummy variables should be created?

- A) 4
- B) 3
- C) 2
- D) 1

Exercise 2 – One-Way ANOVA (5 pts)

Three different teaching methods are tested to evaluate their effect on student exam performance. The following exam scores were obtained. For this exercise, use a significance level $\alpha = 0.05$.

Method A	Method B	Method C
78	72	69
82	74	70
80	76	68
79	73	72
85	75	71
81	77	67

Assume: observations are independent, exam scores are normally distributed, and population variances are equal.

- A) State the null and alternative hypotheses. (1 pt)
- B) Perform a one-way ANOVA test to determine whether the three teaching methods have the same mean exam score. You should compute the group means, the overall mean, the ANOVA quantities, and the F-test statistic (Table A), then state your conclusion at significance level $\alpha = 0.05$. (3 pts)
- C) State your statistical decision and interpret the result. (1 pt)

Useful information for exercise 2

$$F = \frac{MSB}{MSW}$$

$$MSB = \frac{SSB}{df_B}, \quad MSW = \frac{SSW}{df_W} \quad SSB = \sum n_i(\bar{x}_i - \bar{x})^2 \quad SSW = \sum \sum (x_{ij} - \bar{x}_i)^2$$

Exercise 3 – Linear regression (4 pts)

A study examines the relationship between the number of hours studied (x) and the score obtained on a statistics test (y). The following data were collected from 6 students:

Student	Hours (x)	Score (y)
1	2	50
2	3	60
3	5	70
4	6	75
5	8	85
6	10	95

A) Compute the estimated slope b_1 and intercept b_0 of the simple linear regression model **(1 pt)**

B) Using the estimated regression equation, predict the test score for a student who studied **7 hours (1 pt)**.

C) Does the equation that you developed in part (a) provide a good fit for the observed data? Explain. **(2 pts)**

Useful information for exercise 3

$$b_1 = \frac{S_{xy}}{S_{xx}} \quad S_{xx} = \sum (x_i - \bar{x})^2$$

$$S_{xy} = \sum (x_i - \bar{x})(y_i - \bar{y})$$

$$R^2 = 1 - \frac{SSE}{SST} = \frac{SSR}{SST}$$

SSE = sum of squared errors (residual), SST = total sum of squares (total variation), SSR = sum of squares due to regression (model/explained variation).

Exercise 4 – Multiple linear regression (5 pts)

A real-estate agency wants to predict the selling price of an apartment (y, in thousands of euros) based on two variables:

- x_1 = surface area of the apartment (in m^2)
- x_2 = distance to the nearest metro station (in minutes on foot)

A sample of $n = 6$ apartments was collected. The data and some pre-computed values are given below:

Apt.	x_1 (m^2)	x_2 (min)	y (k€)
1	35	3	180
2	50	8	210

3	60	5	250
4	45	12	185
5	80	2	320
6	55	10	215

A) Compute the vector of estimated coefficients. **(1 pt)**

$$X'X = \begin{pmatrix} 6 & 325 & 40 \\ 325 & 18775 & 2055 \\ 40 & 2055 & 346 \end{pmatrix}$$

$$X'Y = \begin{pmatrix} 1360 \\ 75325 \\ 9205 \end{pmatrix}$$

$$XY' = \begin{pmatrix} 180 & 210 & 250 & 185 & 320 & 215 \\ 6300 & 10500 & 15000 & 8325 & 25600 & 11825 \\ 540 & 1680 & 1250 & 2220 & 640 & 2150 \end{pmatrix}$$

$$R = \begin{pmatrix} 1 & -0.32 & 0.94 \\ -0.32 & 1 & -0.55 \\ 0.94 & -0.55 & 1 \end{pmatrix} \quad (X'X)^{-1} = \begin{pmatrix} 4.708 & -0.063 & -0.172 \\ -0.063 & 0.001 & 0.001 \\ -0.172 & 0.001 & 0.015 \end{pmatrix}$$

B) Test whether there is a significant linear relationship between the dependent variable and the explanatory variables at the 5% significance level. **(1.5 pt)**

C) An apartment has a surface area of 58 m² and is located 6 minutes from a metro station. Predict its selling price using your fitted model **(0.5 pt)**

D) The actual selling price of this apartment was 238 000 €. Compute the residual. Is the model over- or under-estimating? What could explain the difference? **(1 pt)**

E) In your opinion, is the sign of β_2 (distance to metro) economically reasonable? Briefly justify your answer. **(1 pt)**

Useful information for exercise 4

$$F = \frac{SSR/df_R}{SSE/df_E}$$

Exercise 5 – Multiple linear regression with categorical data (3.5 pts)

A restaurant review company studies the relationship between overall customer satisfaction y and the following variables:

- x_1 : average meal price (in dollars)
- Restaurant type (French or Seafood/Steakhouse)

A reduced dataset of 6 restaurants is given below:

Rest.	x_1 (\$)	x_2 (type)	y (%)
1	18	French	84
2	17	French	81
3	18	USA	77
4	23	French	86
5	17	USA	75
6	19	USA	81

A) Develop the estimated regression equation showing how overall customer satisfaction is related to the average meal price **(1 pt)**

B) At the 5% significance level, test whether there is a significant linear relationship between average meal price and customer satisfaction. **(1 pt)**

C) Develop the estimated regression equation showing how overall customer satisfaction is related to the restaurant type. **(1 pt)**

D) If you use the customer satisfaction and the restaurant type, the customer satisfaction is modelled by:

$$\hat{y} = 55.80 + 1.10x_1 + 3.20x_2$$

where x_2 is 1 for French restaurant, 0 for US.

Is type of restaurant a significant factor in overall customer satisfaction? **(1pt)**

E) Predict the customer satisfaction score for a USA restaurant that has an average meal price of \$20. How much would the predicted score have changed for a French restaurant? **(0.5pt)**

Useful information for exercise 5

- $t_{crit}(0.05, df=3) \approx 3.182$
- $t_{crit}(0.05, df=4) \approx 2.776$

$$t = \frac{b_1}{SE(b_1)}$$

$$SE(b_1) = \sqrt{\frac{MSE}{\sum (x_i - \bar{x})^2}}$$

$$F = \frac{(SSE_R - SSE_F)/(df_R - df_F)}{SSE_F/df_F}$$

Exercise 6 – Interpreting a Multiple Regression Output (2 pts)

A data analyst at a food delivery company wants to predict the weekly revenue (y , in k€) of delivery drivers based on the following 4 predictors:

Variable	Description
X1	Number of deliveries completed per week
X2	Average distance per delivery (in km)
X3	Customer satisfaction score (out of 10)
X4	Number of years of experience as a driver

SUMMARY OUTPUT

Regression Statistics

Multiple R	0.904163
R Square	0.817511
Adjusted R Square	0.787498
Standard Error	1.832047
Observations	30

ANOVA

	df	SS	MS	F	Significance F
Regression	4	365.8724	91.4681	27.2521	3.18E-09
Residual	25	83.9276	3.3571		
Total	29	449.8000			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%	Upper 95%
Intercept	-4.2731	2.9140	-1.4662	0.1549	-10.2751	1.7289
X1 (Deliveries)	0.1823	0.0241	7.5643	0.0001	0.1328	0.2318
X2 (Avg. Distance km)	-0.0714	0.1052	-0.6788	0.5035	-0.2880	0.1452
X3 (Satisfaction score)	1.4056	0.3817	3.6826	0.0011	0.6196	2.1916
X4 (Years experience)	0.0387	0.1204	0.3215	0.7508	-0.2092	0.2866

A) Does the 95% confidence interval for X2 include zero? Explain what this implies about the statistical significance of X2. **(0.5 pt)**

B) Based on the regression output, is there evidence of a relationship between the dependent variable and the explanatory variables? Explain **(1 pt)**

C) Are all individual coefficients statistically significant? Explain your answer. **(0.5 pt)**

Appendix – Statistical Tables

Table A. F critical values for $\alpha = 0.05$

Upper-tail critical values $F_{0.05, v_1, v_2}$. Here v_1 is the numerator degree of freedom and v_2 is the denominator degree of freedom.

$v_2 \setminus v_1$	1	2	3	4	5	6
5	6.61	5.79	5.41	5.19	5.05	4.95
10	4.96	4.10	3.71	3.48	3.33	3.22
15	4.54	3.68	3.29	3.06	2.90	2.79
20	4.35	3.49	3.10	2.87	2.71	2.60